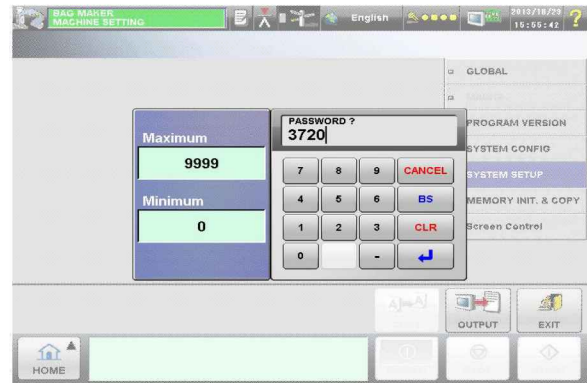


7. Press the [SYSTEM SETUP] index.

►The keyboard to enter the password will appear.

8. Enter the password "3720".

►The SYSTEM SETUP screen will appear.



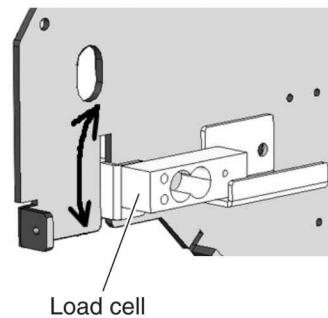
9. Confirm that the following values have been displayed for the load cell data on the screen:

If dancer roller is at lower limit: 520 ( $\pm 10$ )

If dancer roller is at upper limit: 4000-4400

#### NOTE

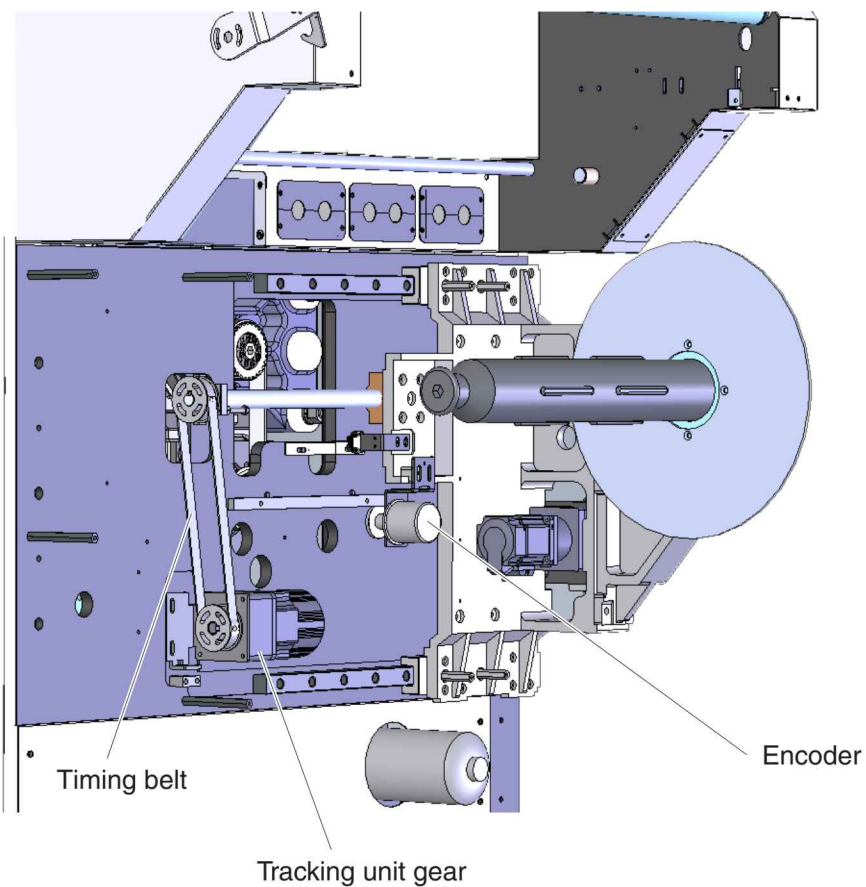
- If the specified values are not given, change the angle of the load cell for adjustment.



10. Loosen the fixing screws, then loosen the bolts from the inside to adjust the position.

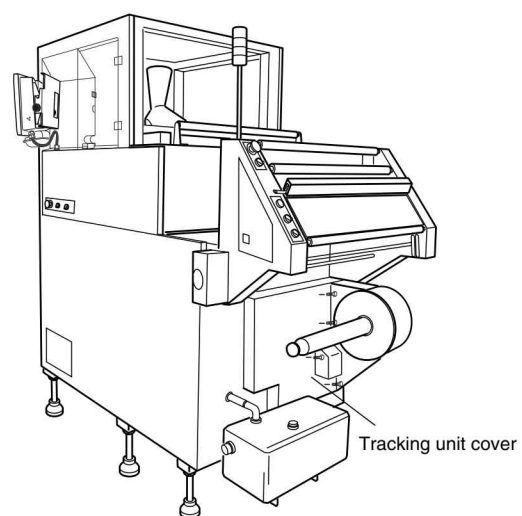
11. Attach the dancer roller cover.

## 4.5 Tracking Unit

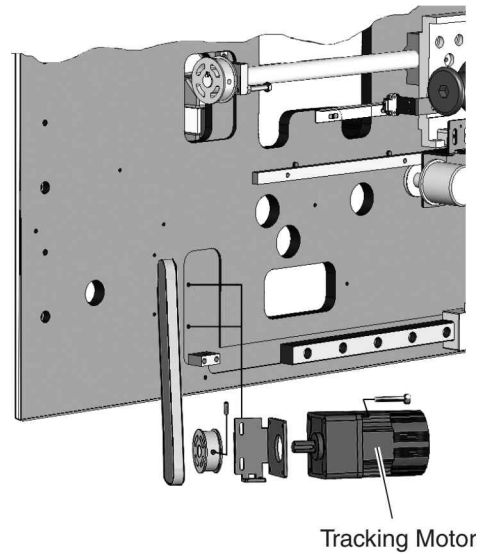


### 4.5.1 M7 Tracking Motor

1. Turn the main power OFF.
2. Remove the back side cover.
3. Loosen 4 bolts of the motor bracket.
4. Loosen the tension of the tracking motor.

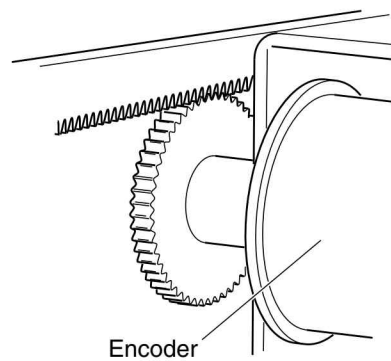


5. Remove the timing belt.
6. Remove the motor bracket by removing 4 bolts.
7. Remove the head gear from the motor.
8. Remove 4 bolts and remove M7 motor from the bracket.
9. Attach the head gear to the new motor.
10. Install the motor with attached head gear.
11. Reinstall all parts in the reverse order of removal.



### 4.5.2 Replacing the Encoder

1. Turn the main power OFF.
2. Remove the back side cover.
3. Remove the two screws and remove the encoder with bracket.
4. Remove the encoder gear.
5. Remove the 4 screws and remove the encoder from the bracket.



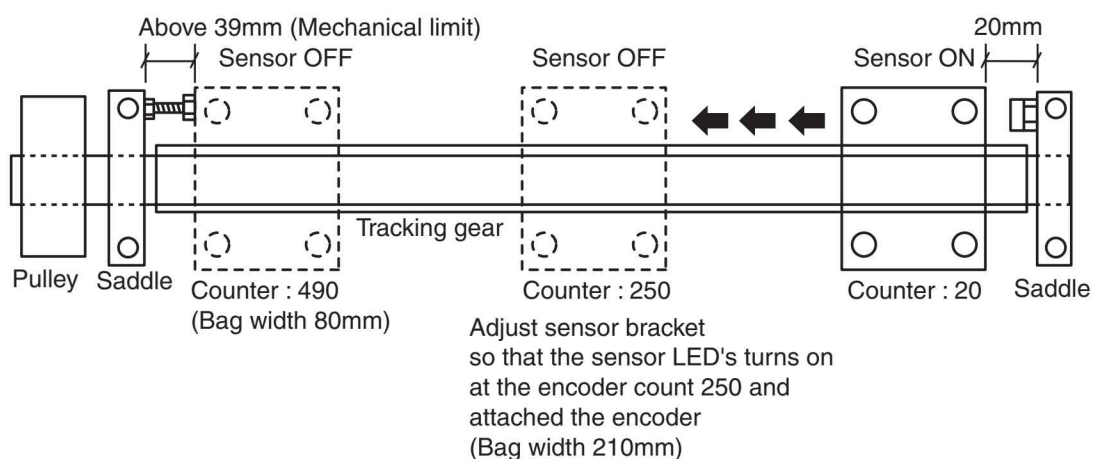
### 4.5.3 Tracking unit Adjustment

#### NOTE

- If you need replacement, replace the parts according to the parts list.

#### Tracking unit Adjustment

- Turn the main power supply ON.
- Remove the Tracking unit cover.
- Switch to the remote control screen's tracking adjustment.
- Set the encoder to count "20" at 20mm from the saddle without pulley.
- Move the tracking unit with rotating the pulley till the encoder counts 250.
- Adjust L plate so that the tracking sensor turns off at this position.
- Additionally, check if the encoder counts 490 at the position of above 39mm from the saddle which is the next of pulley).



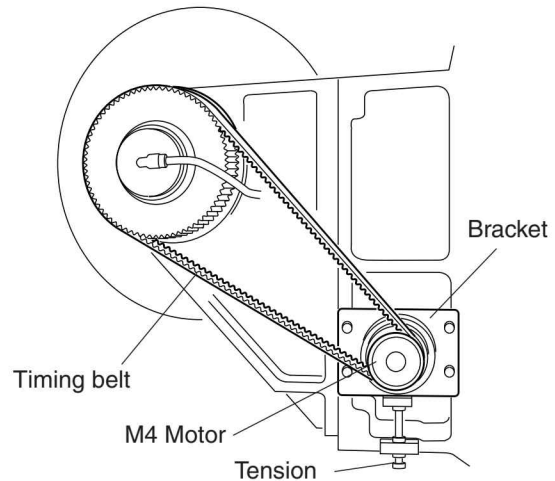
## 4.6 Replacing the Film Unwind Unit

### 4.6.1 Replacing the M4 Motor

1. Turn the main power OFF.
2. Remove the film spindle unit cover.
3. Loosen 4 bolts of the motor bracket, and loosen the tension bolt.
4. Detach the timing belt.
5. Remove 4 screws and remove the motor with bracket.
6. Remove the belt gear.
7. Remove the M4 motor from the bracket.
8. Attach the original gear to the new motor; then apply a small amount of graphite grease.

**NOTE**

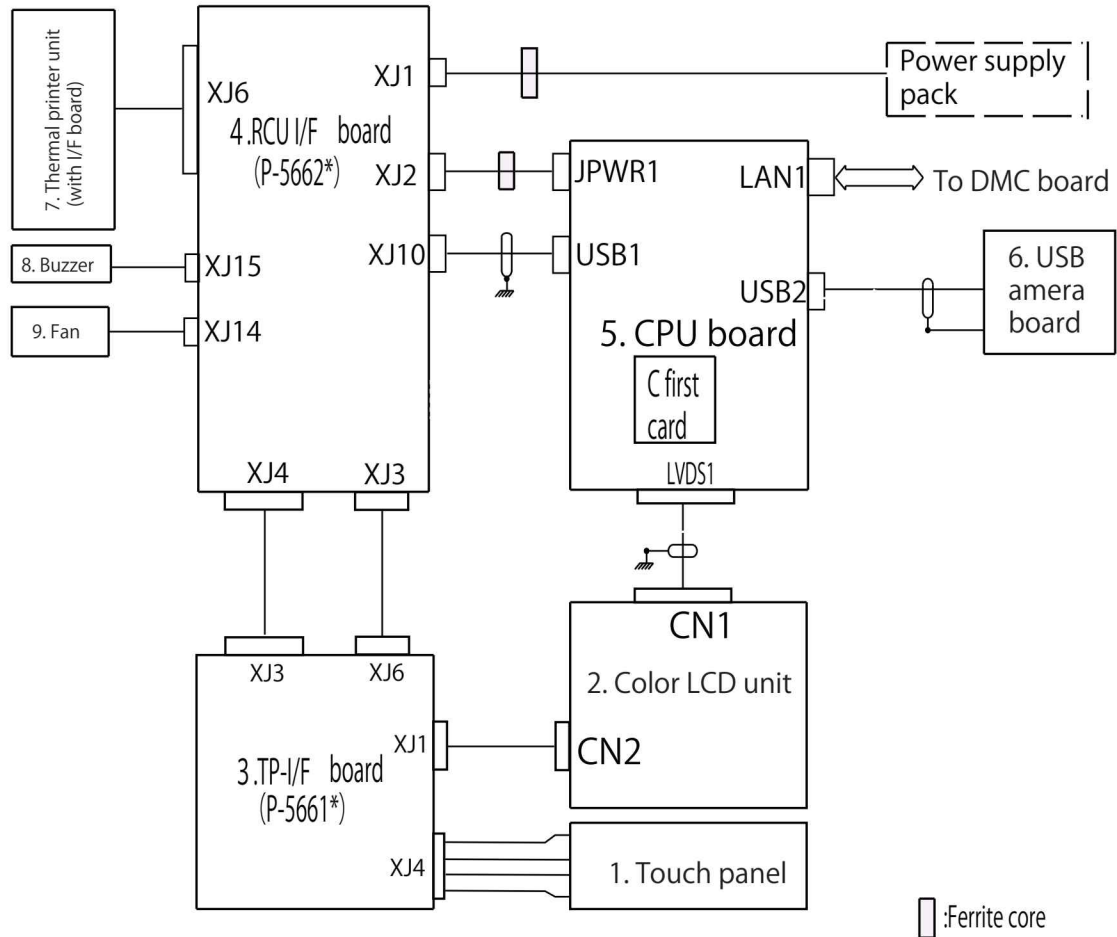
- Replace the motor only. The gears are not included.



9. Install the new motor.
10. Reinstall all parts in the reverse order of removal.

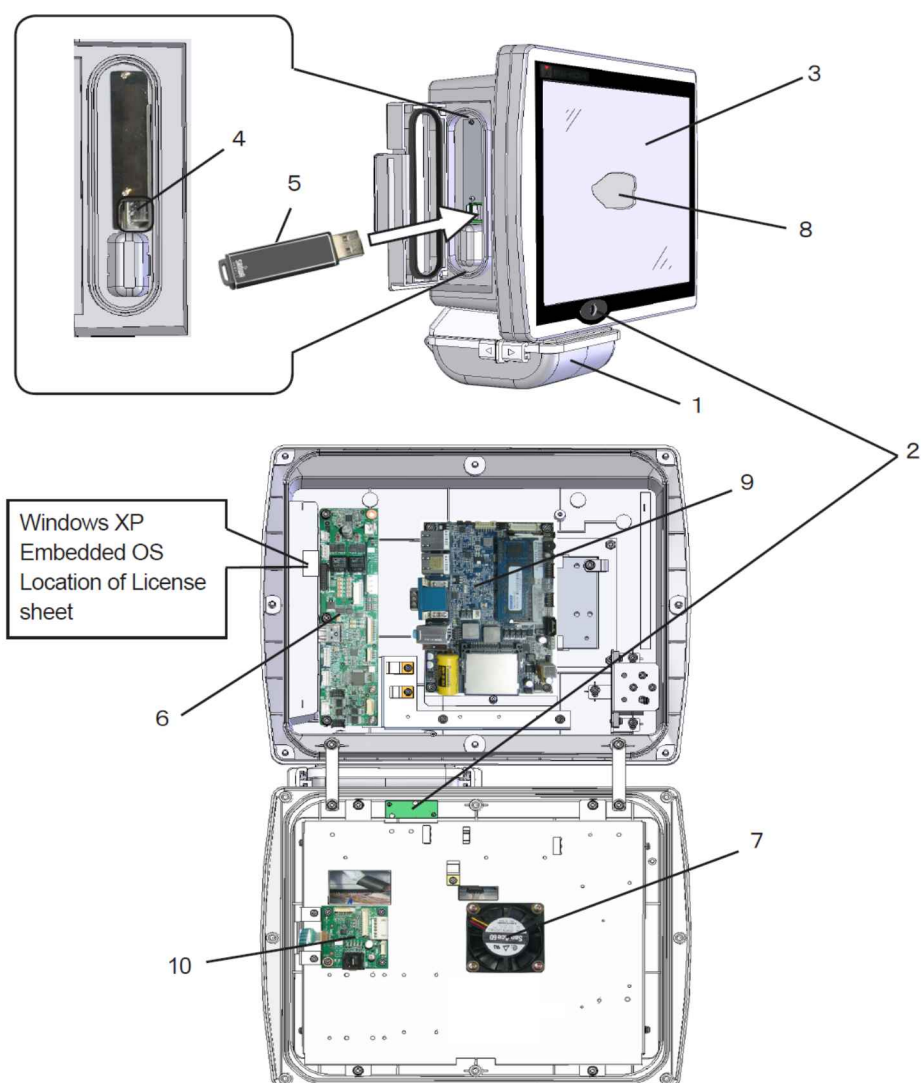
## 4.7 RCU Unit (option)

### 4.7.1 Remote Control Unit Block Diagram



| Symbol | Description                          | Type no. / Notes | Maker      |
|--------|--------------------------------------|------------------|------------|
| 1      | Touch panel                          | KBT-12.1C10R-FM  | KB Elctron |
| 2      | Color LCD unit                       | AC121SA01        | Mitsubishi |
| 3      | TP-I/F board (P-5661*)               |                  |            |
| 4      | RCU-I/F board (P-5662*)              |                  |            |
| 5      | CPU board                            | NPC-M0103        | Omron      |
| 6      | USB camera board                     |                  |            |
| 7      | Printer                              | SAM-1245-10K     | SEIKO      |
| 8      | Buzzer                               |                  |            |
| 9      | Fan                                  |                  |            |
| 10     | USB camera switching board (P-5582*) | (option)         |            |

### 4.7.2 Remote Control Unit Outline View



| No. | Unit Description              |
|-----|-------------------------------|
| 1   | Printer unit                  |
| 2   | USB camera                    |
| 3   | Touch panel                   |
| 4   | USB memory slot (2 slots)     |
| 5   | USB memory                    |
| 6   | RCU I/F board (P-5662*)       |
| 7   | Fan motor                     |
| 8   | Color LCD unit                |
| 9   | CPU board (NPC-M0103) (OMRON) |
| 10  | TP-I/F board (P-5661*)        |



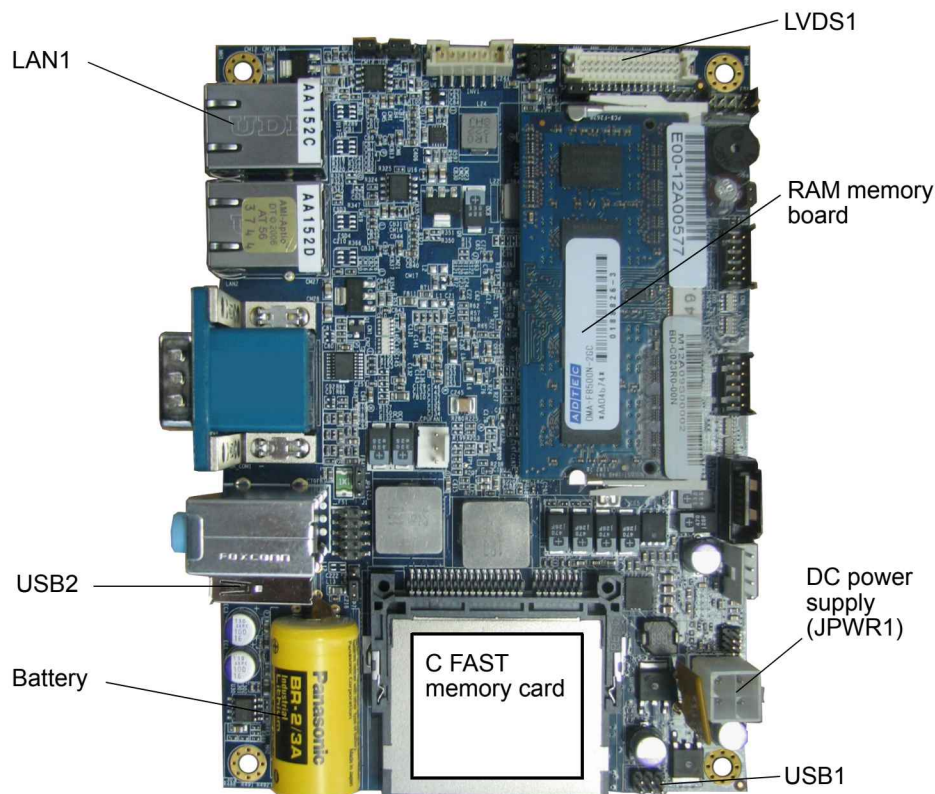
### 4.7.3 CPU Board (NPC-M0103)



#### WARNING

- Before replacing electrical components, make sure that the main power supply and motor power supply are turned OFF. Failure to do so can result in electrical shock.

#### Outline and Connector location



#### Function of board

1. Input control by touch panel
2. Communication [Ethernet] with CAL [DMU board (P-5562 \*)]
3. Communication [USB1 slot] with RCU I/F board (P-5562\*)
4. Controls USB camera, USB camera selection board (P-5562\*)

#### Battery

1. Battery in use: Lithium primary battery (accumulator battery) Model: BR-2/3A
2. Replacement time: 15 years (It depends on the environment.)



**CAUTION**

- The battery replacement with a wrong one may cause malfunction in the board. Replace with the same type or equivalent.
- Discard the used battery.

**Connector classification**

| Connector No. |       | Function                                            |
|---------------|-------|-----------------------------------------------------|
| LAN1          | (8P)  | Ethernet communication (to DMU board)               |
| LVDS1         | (30P) | LCD control                                         |
| JPWR1         | (4P)  | DC power supply (+12 V)                             |
| USB1          | (6P)  | USB communication (to RCU I/Fboard)                 |
| USB2          | (4P)  | Control of USB camera,<br>USBcamera switching board |

**Connector function in details**

- (1) JPWR1(DC-IN): Power supply input

| Connector No.    | Pin No. | Signal name | Remarks |
|------------------|---------|-------------|---------|
| DC-IN<br>(JPWR1) | 1       | GND         |         |
|                  | 2       | GND         |         |
|                  | 3       | +12 V       |         |
|                  | 4       | +12 V       |         |

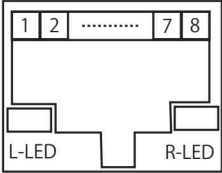
- (2) USB1:USB (internal connection) interface

| Connector No.                                       | Pin No. | Signal Name   | Pin No. | Signal Name |
|-----------------------------------------------------|---------|---------------|---------|-------------|
| U S B 1<br>Interface<br>(forInternal<br>connection) | 1       | V C C _ U S B | 2       | G N D       |
|                                                     | 3       | U S B O N     | 4       | G N D       |
|                                                     | 5       | U S B O P     | 6       | ( N . C . ) |

## (3) USB2:USB (external connection) interface

| Connector No.                                        | Pin No. | Signal name   | Remarks |
|------------------------------------------------------|---------|---------------|---------|
| U S B 2<br>Interface<br>(for External<br>connection) | 1       | V C C _ U S B |         |
|                                                      | 2       | U S B N       |         |
|                                                      | 3       | U S B P       |         |
|                                                      | 4       | G N D         |         |

## (4) LAN1: LAN interface

| Connector No.                                                                                                  | Pin No. | Signal name | Remarks |
|----------------------------------------------------------------------------------------------------------------|---------|-------------|---------|
| L A N 1<br>Interface<br><br> | 1       | MDI_OP      |         |
|                                                                                                                | 2       | MDI_ON      |         |
|                                                                                                                | 3       | MDI_1P      |         |
|                                                                                                                | 4       | MDI_2N      |         |
|                                                                                                                | 5       | MDI_2P      |         |
|                                                                                                                | 6       | MDI_1N      |         |
|                                                                                                                | 7       | MDI_3P      |         |
|                                                                                                                | 8       | MDI_3N      |         |

## L-LED

|             |       |
|-------------|-------|
| Green : ON  | 100 M |
| Orange : ON | Giga  |
| OFF         | 10 M  |

## R-LED

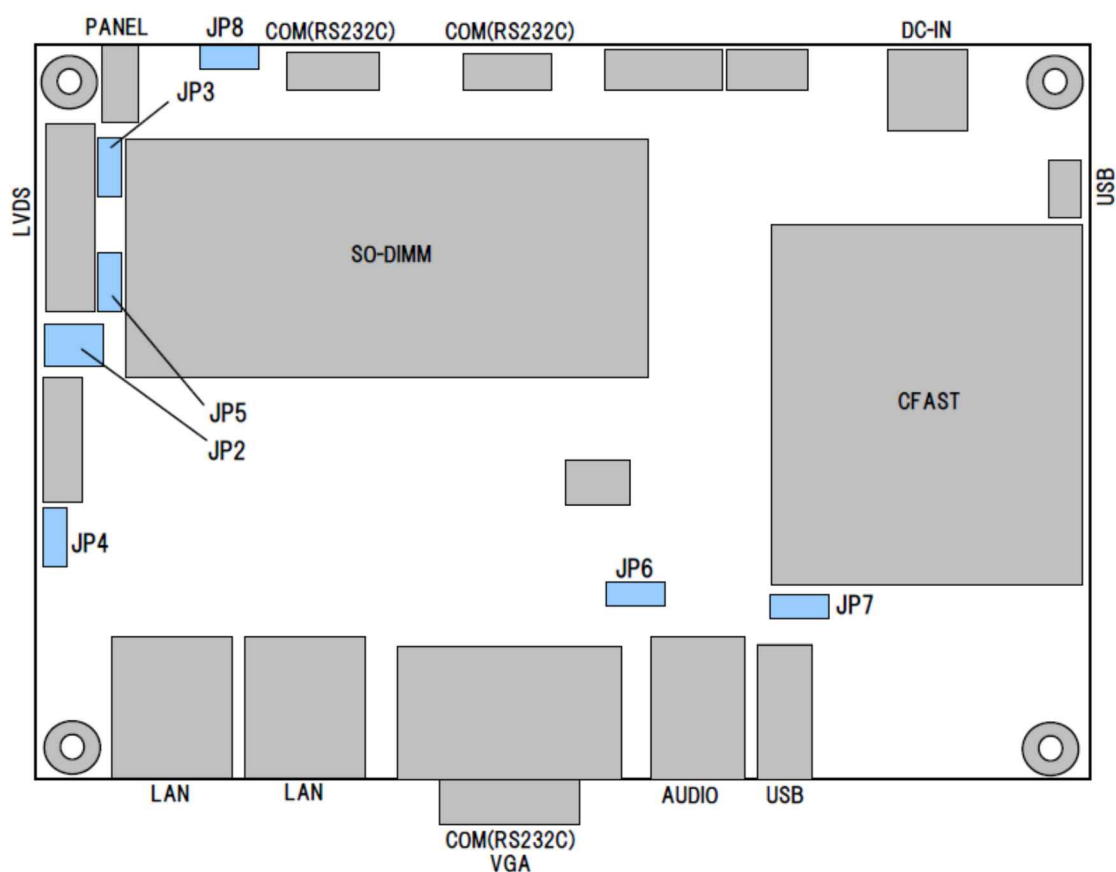
|                |      |
|----------------|------|
| Yellow : blink | Act  |
| Yellow : ON    | Link |
| OFF            | OFF  |

## (5) LVDS1: LCD interface

| Connector No.             | Pin No. | Signal name | Remarks | Pin No. | Signal name | Remarks |
|---------------------------|---------|-------------|---------|---------|-------------|---------|
| LVDS1<br>LCD<br>Interface | 1       | LVDS_VCC    |         | 2       | LVDS_VCC    |         |
|                           | 3       | LVDS_VCC    |         | 4       | LVDS_VCC    |         |
|                           | 5       | GND         |         | 6       | GND         |         |
|                           | 7       | GND         |         | 8       | GND         |         |
|                           | 9       | SELLVDS     |         | 10      | N.C         |         |
|                           | 11      | N.C         |         | 12      | N.C         |         |
|                           | 13      | GND         |         | 14      | GND         |         |
|                           | 15      | GND         |         | 16      | LVDS0_CLK+  |         |
|                           | 17      | LVDS0_CLK-  |         | 18      | GND         |         |
|                           | 19      | LVDS0_D2+   |         | 20      | LVDS0_D2-   |         |
|                           | 21      | GND         |         | 22      | LVDS0_D1+   |         |
|                           | 23      | LVDS0_D1-   |         | 24      | GND         |         |
|                           | 25      | LVDS0_D0+   |         | 26      | LVDS0_D0-   |         |
|                           | 27      | GND         |         | 28      | LVDS0_D3+   |         |
|                           | 29      | LVDS0_D3-   |         | 30      | GND         |         |

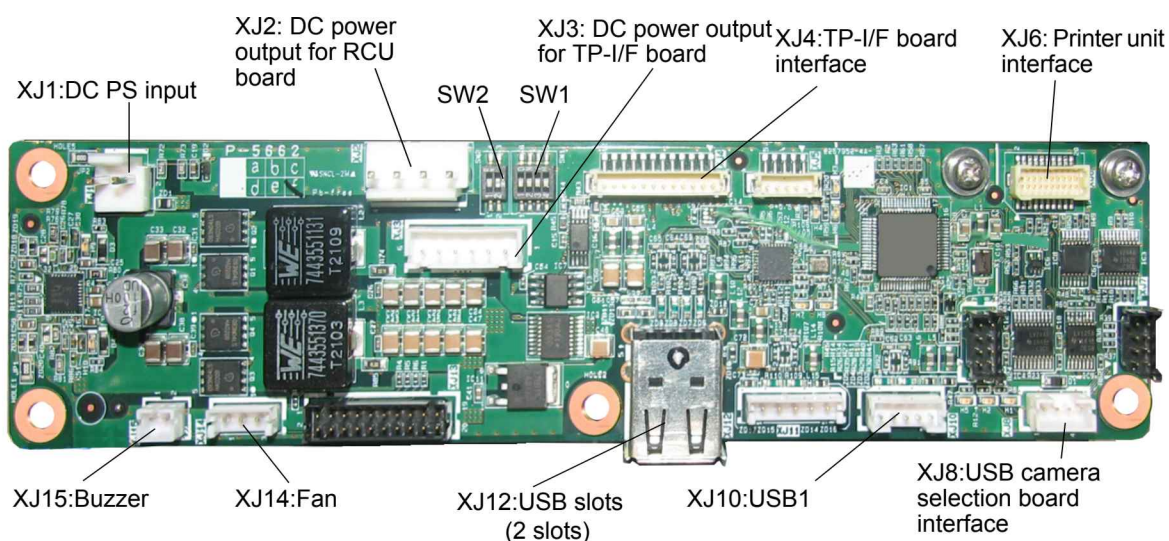
## Junper Setting

| JP No. | Function          | Selection              | Position of jumper | Remarks                                                                         |
|--------|-------------------|------------------------|--------------------|---------------------------------------------------------------------------------|
| J P 2  | LVDS_VCC          | 3.3 V                  | * 1-3 2-4          |                                                                                 |
|        |                   | 5.0 V                  | 3-5 4-6            |                                                                                 |
| J P 3  | SEL LVDS          | High                   | 1-2                | Setting for pin 9 of LVDS connector<br>The voltage level of "High" is LVDS_VCC. |
|        |                   | Low                    | 2-3                |                                                                                 |
| J P 4  | ENABKL            | 3.3 V                  | * 1-2              | Pin 1 of BACKLIGHT connector.<br>Voltage level setting of High.                 |
|        |                   | 5.0 V                  | 2-3                |                                                                                 |
| J P 5  | LVDS0_D3+         | 1.8 V                  | * 1-2              | Setting for pin 29 of LVDS connector.                                           |
|        |                   | 3.3 V                  | 2-3                |                                                                                 |
| J P 6  | Clear CMOS        | Normal                 | * 1-2              |                                                                                 |
|        |                   | Clear CMOS             | 2-3                |                                                                                 |
| J P 7  | C fast PS voltage | 3.3 V                  | * 1-2              |                                                                                 |
|        |                   | 5.0 V                  | 2-3                |                                                                                 |
| J P 8  | Power ON          | Hardware Auto Power ON | * 1-2              | Starts regardless the setting of Restore AC Power Loss of BIOS after power ON.  |
|        |                   | N.C                    | 2-3                |                                                                                 |



## 4.7.4 RCU I/F Board (P-5662\*)

### Appearance, Connector and DIP-SW locations



### Function of board

1. TP-I/F board (P-5561\*) control
2. Power supply and USB communication with CPU board
3. Communication with USB slots
4. Buzzer control
5. Printer control
6. Fan control

### DIP-SW Setting

#### (1) SW1

| SW 1 | Function              | Selection       | ON/OFF | Remarks                |
|------|-----------------------|-----------------|--------|------------------------|
| 1    | Not use               |                 | * OFF  |                        |
|      |                       |                 | ON     |                        |
| 2    | Print mode            | Inverted mode   | * OFF  | Handstand mode at CCW  |
|      |                       | Handstand mode  | ON     |                        |
| 3    | Delay start           | 0 sec           | * OFF  | Delay start for DACS-G |
|      |                       | 45 sec          | ON     |                        |
| 4    | Camera selection mode | Normal          | * OFF  |                        |
|      |                       | Reset at switch | ON     |                        |

\* : Default setting

#### (2) SW2

| SW 2 | Function           | Select       | ON / OFF | Remarks |
|------|--------------------|--------------|----------|---------|
| 1    | Touch panel        | 4 wires      | * OFF    |         |
|      |                    | 5 wires      | ON       |         |
| 2    | Serial output port | XJ7(DF11-8P) | OFF      |         |
|      |                    | XJ6(SHD-20P) | * ON     |         |

\* : Default setting